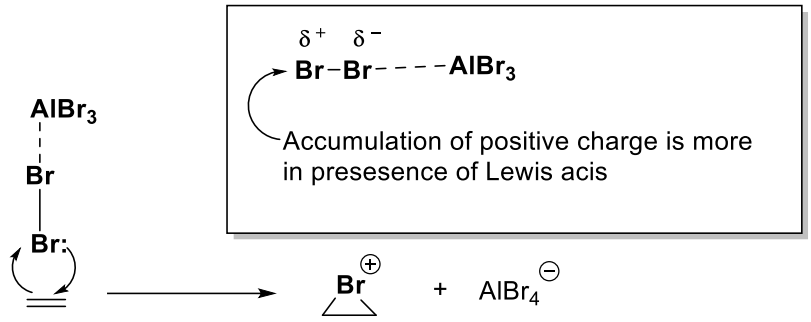
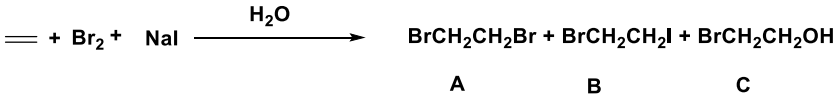
1. The rate-determining step of bromine addition is the electrophilic attack by the positively polarized end of the bromine dipole on the alkene double bond. The greater the polarization of the bromine molecule, the faster the reaction. A Lewis acid like $AlBr_3$ increases polarization of the Br–Br bond, enhancing the positive charge on the near end and accelerating the reaction.



Whereas in case of KBr, halide ion (Br^-) combines with halogen molecules (Br_2) to form complex polyhalide ion Br_3^-. Thus, the addition of KBr inhibits the bromine molecule to act as an electrophile and makes the reaction slow.

2. The experimental evidence in favor of a two-step pathway and against a one-step pathway is as follows:

   i. When addition of bromine is carried out in presence of other nucleophile, in addition to 1,2-dibromide product another product is obtained which contains one bromine atom and another nucleophile atom.

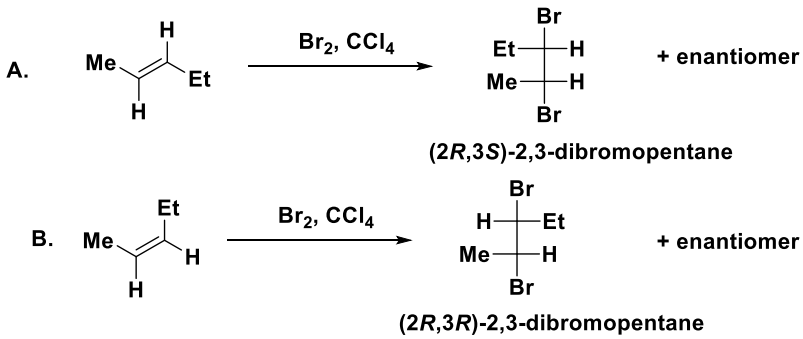


In this example 3 compounds are formed, which clearly shows that it goes via two step pathways. Dihalide B and alcohol C cannot be obtained by direct displacement of bromine atom and if it goes via one step pathway then there is no scope of attack by I^- or H_2O.

   ii. Bromination of alkene which may yield different products through syn and anti-addition is another tool for establishing the mechanistic pathway. For instance, cis-2-butene may give racemic dibromide on whereas trans-2-butene gives meso product. (See class notes)

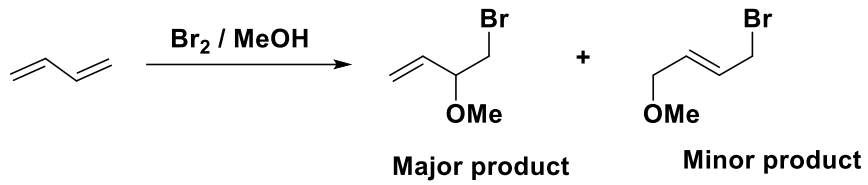
3. Follow class note.

4.



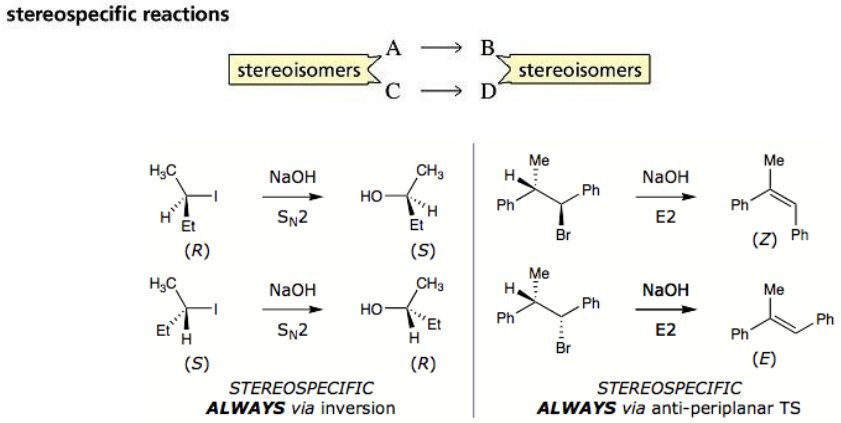
(practice the mechanism)

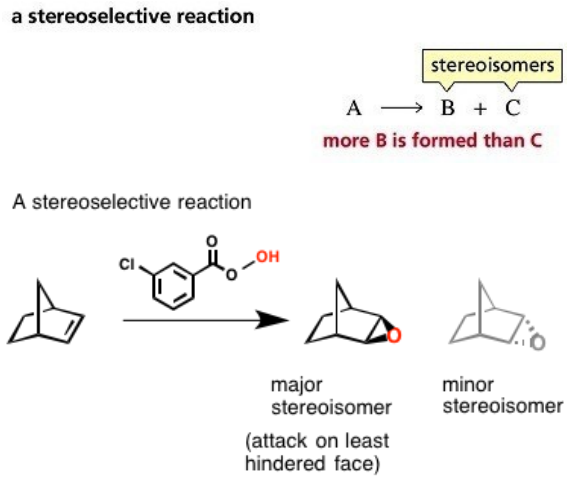
5.



6. Check text book for additional reading.

Stereospecific reactions lead to the production of a single isomer as a direct result of the mechanism of the reaction and the stereochemistry of the starting material. The reaction gives a different diastereoisomers of the product from each stereoisomer of starting material.



In stereoselective reactions one steresoisomeric product is preferred over another.

Stereoselective reactions give one predominant product because the reaction pathway has a choice. Either the pathway of lower activation energy is proffered (kinetically controlled) or the more stable product is preferred (thermodynamically controlled).